

Piyush Kumar

Data Scientist (3.5+ Years) | +91 9708859326 | pyushkumar999@gmail.com | [GitHub](#) | [Portfolio](#)

Objective

Data Scientist with 3.5+ YOE in building, deploying, and optimizing ML models using Python and Azure environments. Seeking to leverage strong expertise in predictive modelling, and MLOps to deliver scalable, business-driven solutions.

Technical Skills

ML / Data Science: *Pandas, PyTorch, TensorFlow, Scikit-learn, NLTK, NLP, Data Structures, Algorithms*

Technologies & Tools: *Azure MLOps, Git, Jenkins, Docker, SQL Server, SDLC, Jira, Confluence, Agile*

Languages: *Python, Javascript, PL/SQL, C++*

Experience

Tata Consultancy Services Ltd., Developer (Kolkata, India) **July 2022 – Present**

Data Scientist (CPD Team, Kolkata) | (Azure MLOps, Python, PyTorch, TensorFlow, AKS) **Sep 2023 – Present**

- Designed and deployed robust end-to-end **MLOps** workflows in Azure ML, leveraging **Azure DevOps** for automated CI/CD pipelines to streamline model training and deployment.
- Built and optimized scalable deep learning models for image classification using **TensorFlow** and **PyTorch**, achieving a 12% accuracy improvement over baseline models.
- Implemented **MLflow** for comprehensive experiment tracking and model registry, which reduced the average model deployment time by 35%.
- Developed and containerized ML services using **Docker**, and managed their scalable deployment on **Azure Kubernetes Services (AKS)** for production environments.
- Automated model monitoring and drift detection post-deployment, enabling retraining triggers within CI/CD pipelines to ensure continuous model performance and reliability.

PL/SQL Developer (Offshore Team, Kolkata) | (Oracle PL/SQL, SQL, Oracle Fusion, SDLC) **July 2022 – Sep 2023**

- Spearheaded the full **SDLC** as an Oracle Developer, architecting high-performance database schemas and optimized indexing strategies that ensured 100% data integrity for critical analytics and reporting.
- Engineered sophisticated **PL/SQL** logic and complex stored procedures to transform large-scale datasets, streamlining business-critical processing and aggregating massive data volumes into actionable insights.
- Partnered with cross-functional stakeholders to translate complex business requirements into robust technical specifications, leveraging Informatica Power Center to manage seamless **ETL** pipelines across Oracle and **SQL Server** environments.

Cognizant Tech. Solutions, Azure DevOps Intern (Kolkata, India)

Mar 2022 – June 2022

Azure DevOps Engineer Intern (Azure Team) | (Azure, Git, Docker, Jenkins, Kubernetes)

- Created and maintained *system documentation*, ensuring compliance with industry standards and improving overall system reliability.
- Implemented **CI/CD** pipelines using **Azure DevOps**, **Jenkins**, and **Docker**, gaining foundational knowledge in **MLOps** for automating the build, testing, and deployment of applications.
- Collaborated with development teams to monitor and analyze model/system performance, a key practice in maintaining production-level machine learning systems.

Projects

Fake News Detection using Machine Learning Algorithm ([Here](#))

- Developed a machine learning model to detect fake news articles. Preprocessed text data using **NLTK**, and extracted features using **TF-IDF**.
- Employed **TF-IDF vectorization** to transform the textual data into numerical features that the model could learn from. Trained a **Logistic Regression** model to classify news articles based on the extracted features.
- Evaluated the model's performance on a test dataset using the accuracy metric and achieved an accuracy score of 96% on the test dataset, demonstrating the model's effectiveness in identifying fake news.
- Tools & Technologies Used:**
 - Tools:** *Pandas, Scikit-learn, NLTK*
 - Technologies:** *Python, Google Colab*

Education and Certifications

Institute of Engineering & Management, Kolkata

B.Tech in Electronics and Communication Engineering

Microsoft Certified: Azure Fundamentals (AZ-900), Azure Data Scientist Associate (DP-100)

July 2022

CGPA – 9.2